

STRUCTURA.AI

“Compressing structural design validation from days to seconds”

Team :

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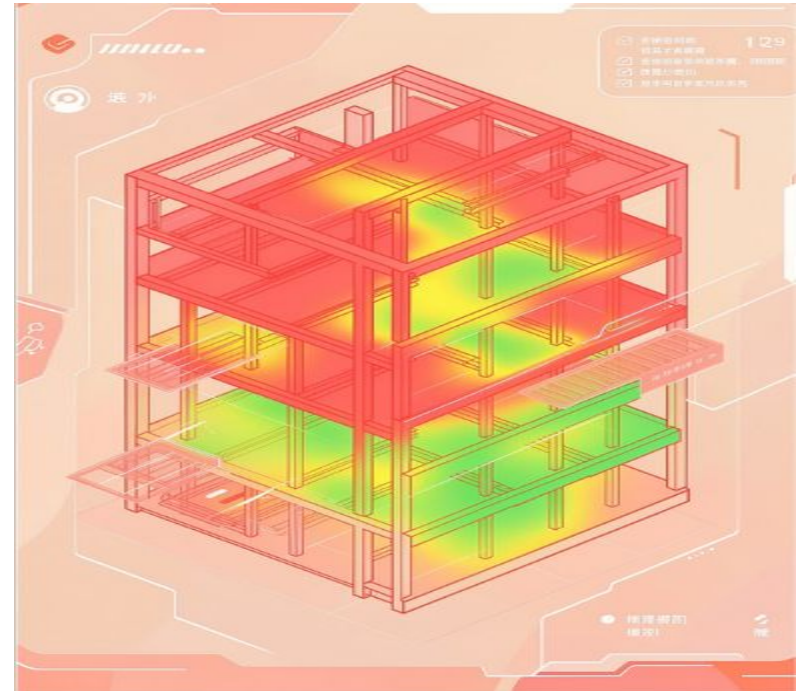
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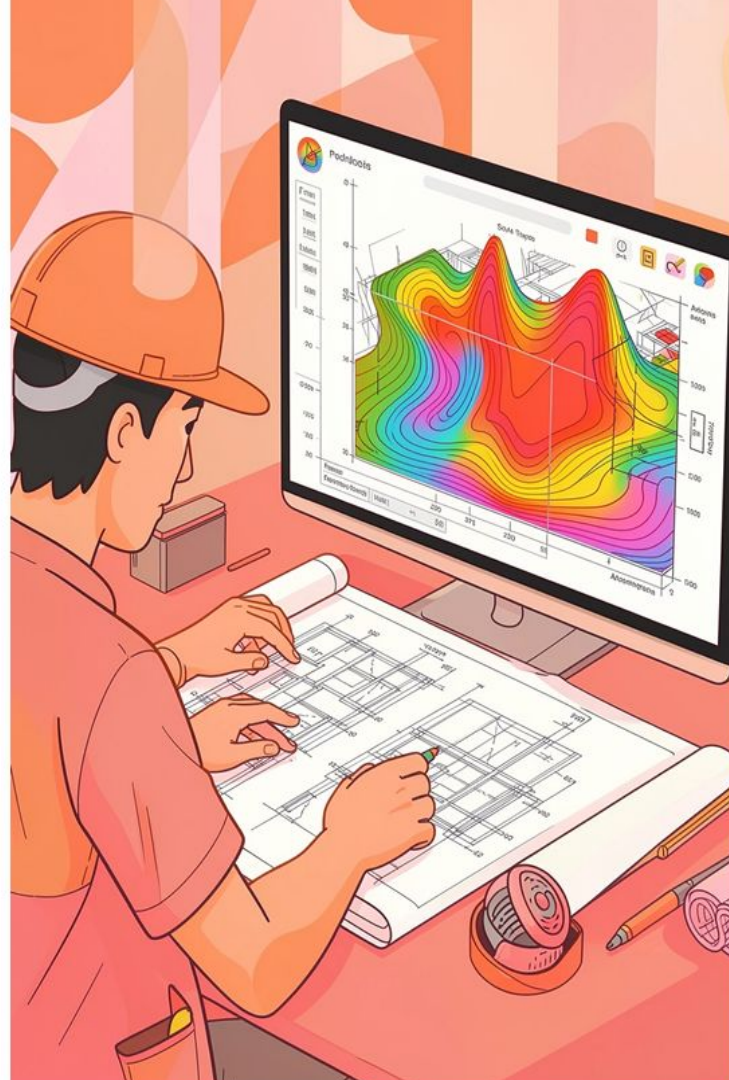


THE CUSTOMER NEED

The Bottleneck: Slow Design Validation

Structural engineering workflows suffer from a costly operational bottleneck between early-stage design and intensive validation. Standard Finite Element Analysis (FEA) workflows using software like ANSYS or ABAQUS are slow, consuming **16 to 48 hours** for moderately complex structures.

Simulations are almost always run **after a design is substantially complete**. When a critical failure point is flagged during FEA, teams must revise the CAD model, remesh, and re-simulate — losing a full cycle of resources. This reactive cycle is expensive, time-consuming, and entirely avoidable.





WHY THIS NEED IS IMPORTANT

Why It Matters: Profit, Safety, and Competitiveness



Direct Cost of Rework

High-cost senior engineers spend days waiting for simulations instead of producing deliverables, crippling firm profitability and billing efficiency.



Pre-Construction Safety Risk

Flaws missed in the design phase can lead to partial or total structural collapse resulting in multi-million rupee litigation or loss of life.



Competitive Disadvantage

Firms are limited to testing only 1 to 3 structural variants, meaning slower turnaround times on RFPs and weaker technical proposals compared to competitors.

Our Solution: Structura.AI

Structura.AI is an instant, web-based structural failure pre-screening platform that sits **upstream** of the traditional FEA pipeline — catching critical risks before they become costly rework.



Rapid Feedback

Engineers upload a 2D layout and basic load parameters, and our deep learning engine returns predictive safety feedback in **under 3 seconds** — not hours.



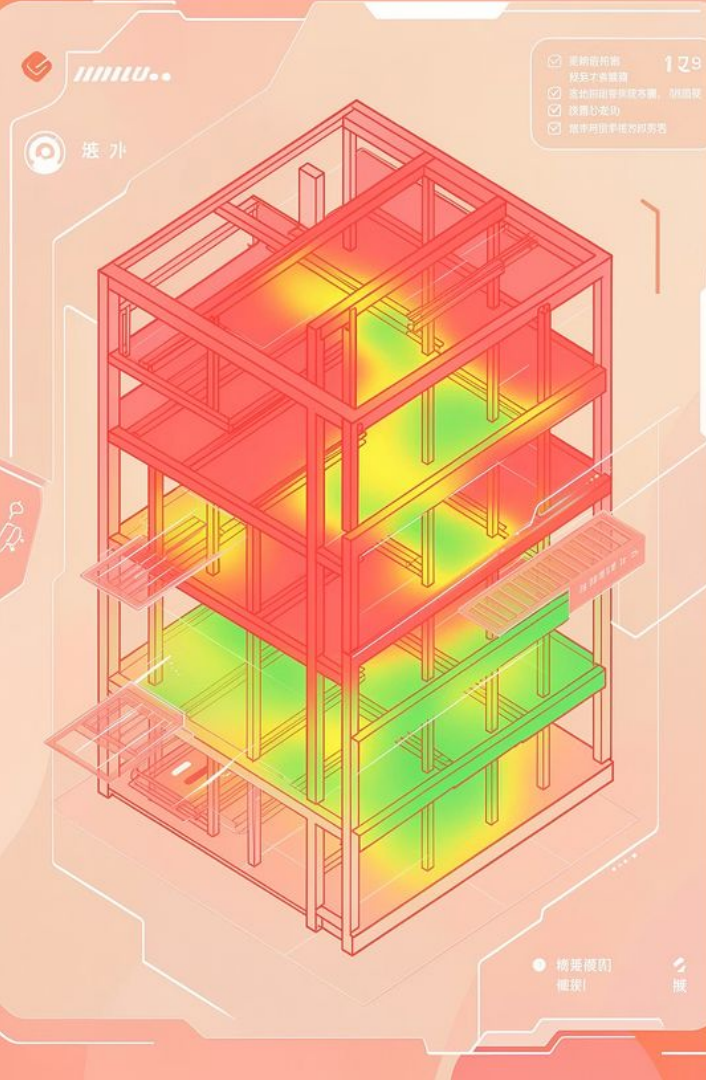
Visual Heat Maps

Provides an instant color-coded overlay — **Red, Yellow, Green** — identifying zones of predicted stress concentration and immediate failure risks before formal simulation begins.



Smart Integration

Structura.AI does **not replace** formal FEA for regulatory sign-off. It refines designs rapidly so that formal simulation becomes a **confirmation step**, eliminating costly late-stage rework.



How **Structura.AI** Solves the Bottleneck

Traditional FEA is a **late-stage bottleneck** — 6–10 hours per iteration, catching failures too late. Structura.AI inserts an **AI-powered pre-screening layer** upstream, compressing the entire cycle.

3s

Predictive Assessment

Replaces a 6–10 hour FEA iteration

10x

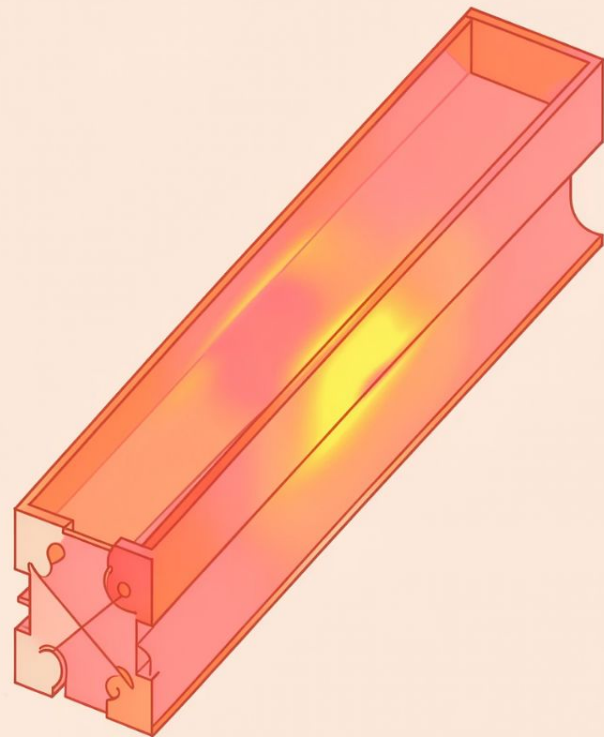
Iteration Speed

50+ design variants vs. 1–3 before

0

Late-Stage Rework

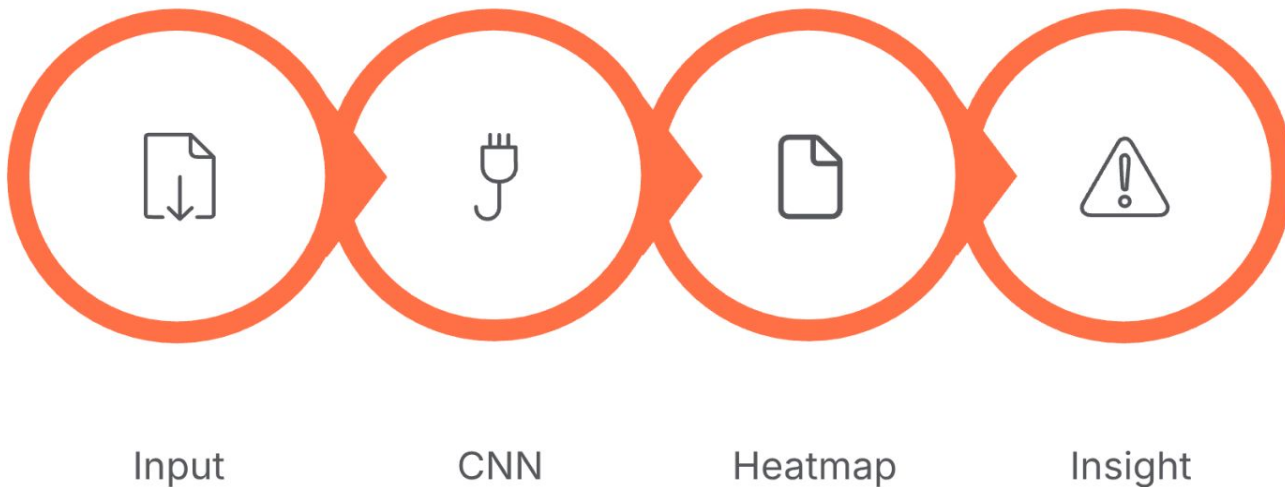
Formal FEA becomes confirmation, not
discovery



Under the Hood: **Structura.AI** Technology

A web-based **Convolutional Neural Network (CNN)** trained on

pre-run FEA simulations — delivering visual failure predictions in seconds, not hours.



The pipeline accepts standard **DXF or JSON** geometry with user-specified loads, processes them through the CNN, and outputs a **color-coded 2D heatmap** overlaid on the structural diagram — **red** for immediate failure risk, **green** for structurally sound zones.

Reaching Our Customers

Three channels spanning the full engineer lifecycle — from student to enterprise decision-maker.



CAD Software Plugins — Primary

Native integrations into **AutoCAD** and **Revit** let engineers run pre-checks directly in their drafting environment — zero workflow disruption.



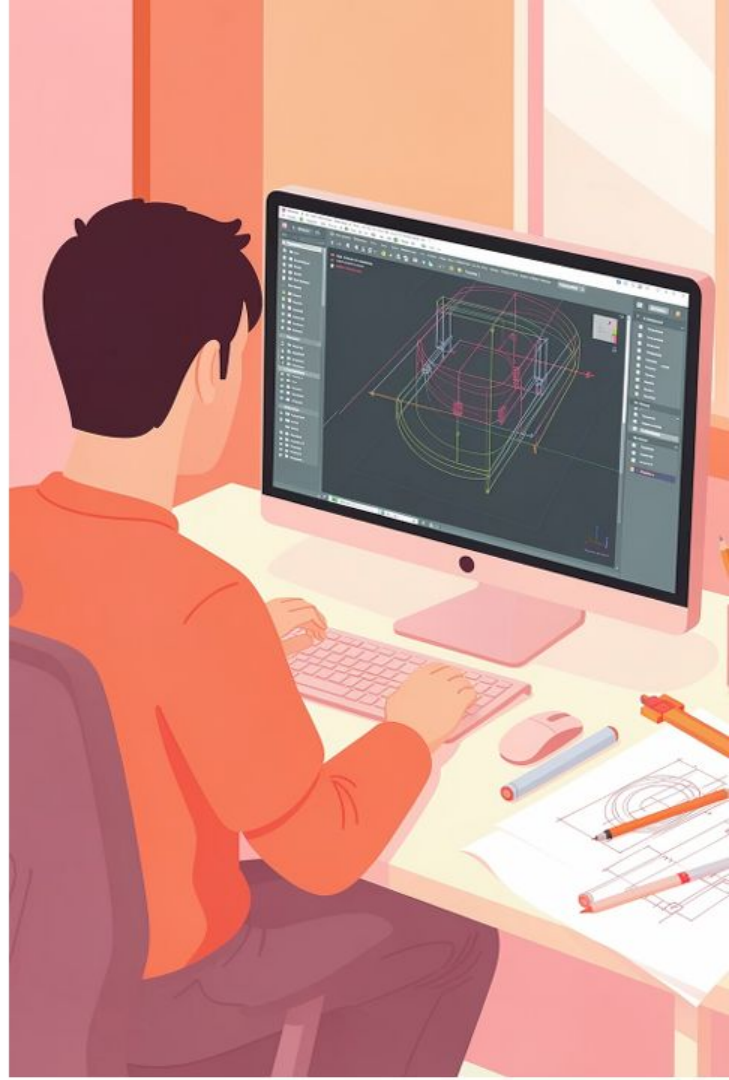
Direct B2B Enterprise Sales

Targeting **Principal Engineers** and **Technical Directors** at tier-1 consultancies. The pitch centers on direct ROI: recovered billing margins and saved engineering hours.



Engineering University Partnerships

Offered at **no cost** to students, building long-term brand equity and embedding Structura.AI into the workflows of tomorrow's industry professionals.



01

Cost-Effective Reach

How we reach customers without burning cash

~₹0

CAC for university channel

1

CAD Plugin Integration

~₹0 incremental CAC

Ship Structura.AI as a free plugin for AutoCAD / Revit. Engineers discover it inside their daily workflow — no paid ads, no cold calls.

2

Direct B2B Demo Video

One-time ~₹5,000 effort

90-sec screen-recorded demo: Structura.AI in 3 sec vs. ANSYS in 10 min. Sent to 50 target firms by email. High signal, near-zero cost.

3

University Partnerships

₹0 — free student tier

Students trained on Structura.AI become advocates inside future employer firms. Builds a word-of-mouth pipeline with zero marketing spend.

Sales & Usage Projections

Next 3 months → End of Year 1

3 MONTHS

Seed Pilot Phase

3–5

Pilot Firms

structural consultancies

15–25

Active Users

senior engineers (free pilot)

₹0–₹30K

Revenue

discounted/free trials

500+

Iterations Tested

per firm, replacing FEA loops



YEAR 1

Growth & First Revenue

12–18

Client Firms

consultancies + EPC firms

60–90

Paid Seats

@ ₹6,000–₹10,000/seat/month

₹4–9 L

Monthly Revenue

by month 12 (MRR)

₹50–80 L

ARR Run Rate

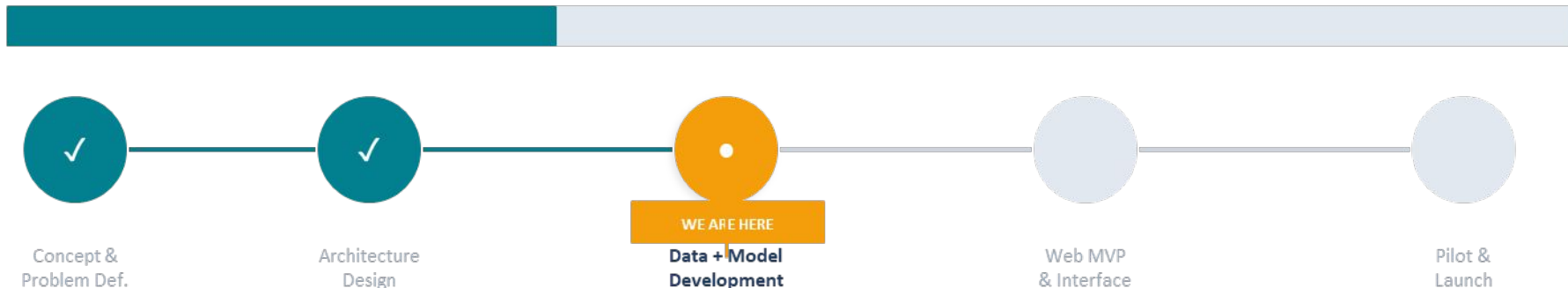
end of year 1 target

Development Progress

Where we are — product & business

Overall Readiness

35%



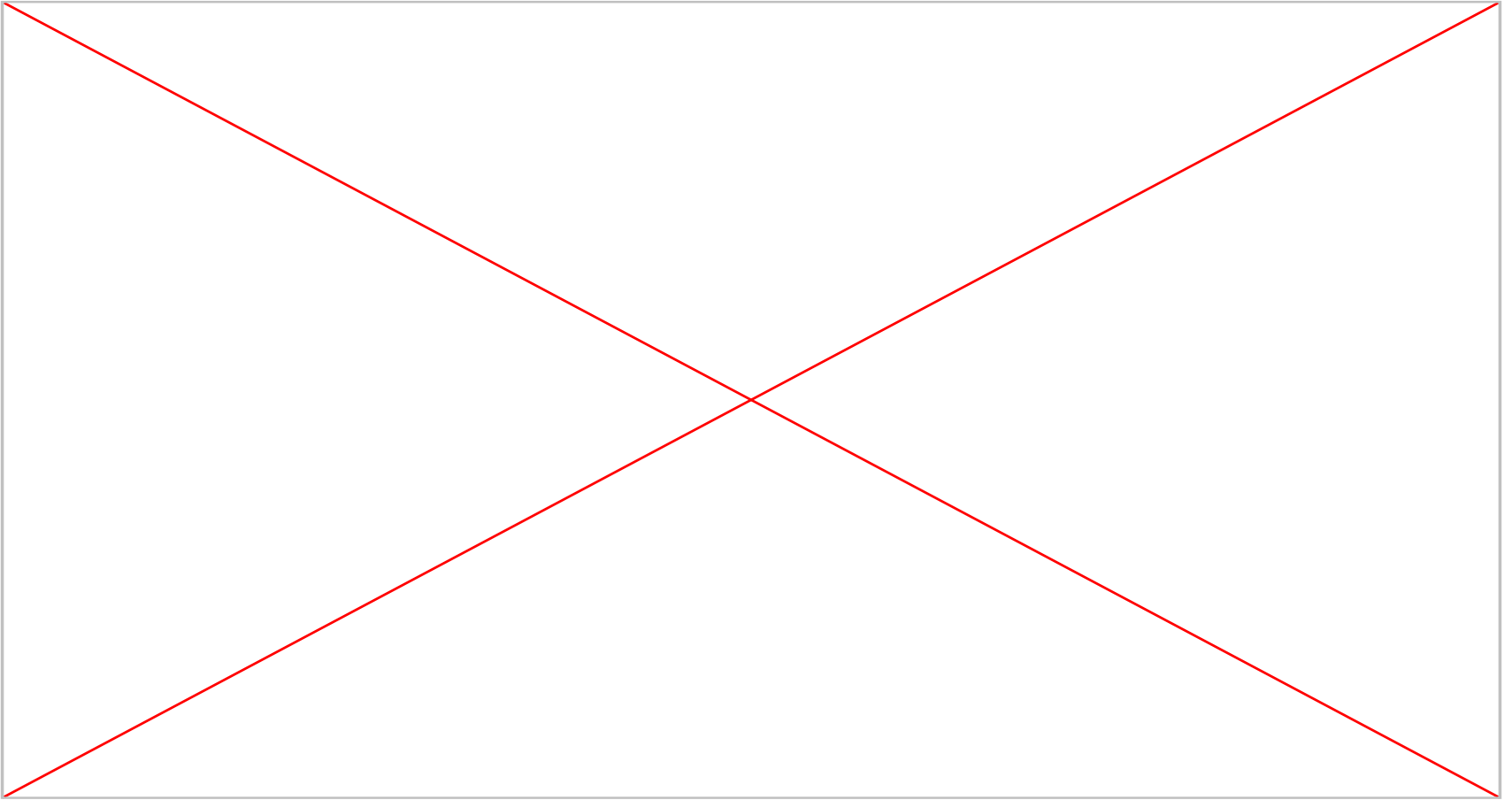
PRODUCT

- ML pipeline architecture fully mapped
- Baseline CNN in PyTorch — being implemented
- FEA dataset curation underway (target 2,000+ configs)
- Model training & 85% accuracy validation — Week 2
- Web UI (Flask/FastAPI) — Week 3 milestone

BUSINESS

- Problem validated — FEA rework cost data documented
- 3 target customer segments identified
- B2B demo video planned (Week 3 deliverable)
- 3 structured engineer interviews — Week 2
- Pilot firm outreach — starting Week 3

Working prototype



Website

